EXP NO: 6 **DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE**

Aim:

To design Class Diagram and Sequence Diagram for the given project.

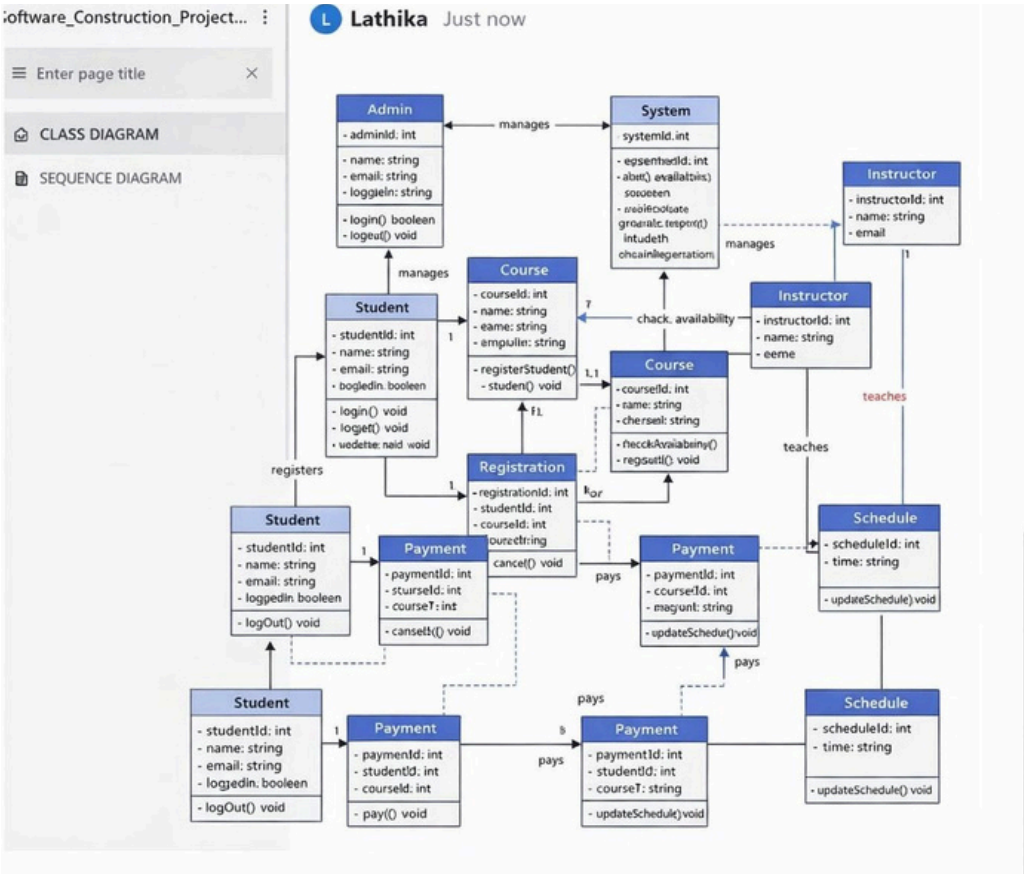
Procedure:

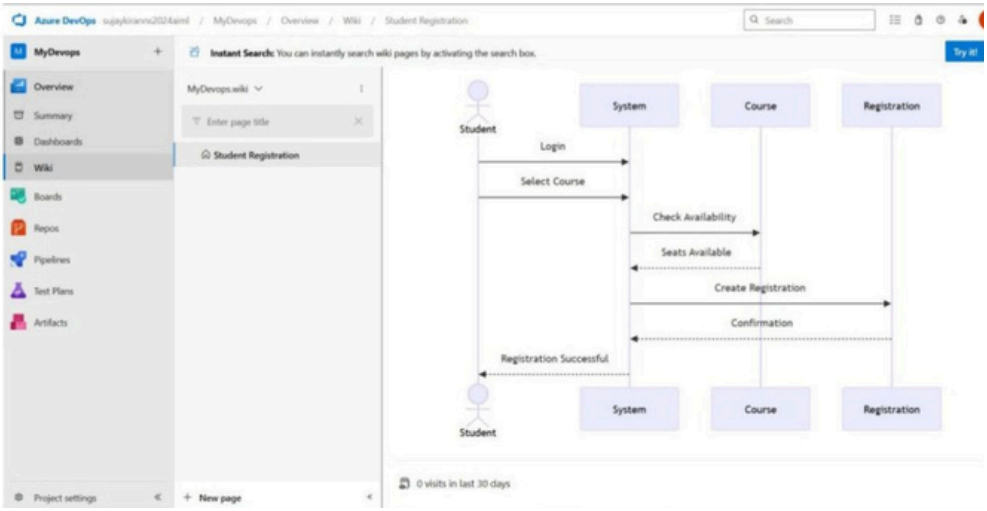
6A. Class Diagram:

1. **Identify main classes:**
   - **User**
   - **Playlist**
   - **Song**
   - **BatchProcessor**
2. **Define attributes for each class:**
   - **Example:**
     - **User → userID, name, email**
     - **Playlist → playlistID, title**
3. **Define methods:**
   - **Example:**
     - **createPlaylist()**
     - **addSong()**
     - **processBatch()**
4. **Establish relationships:**
   - **Association (User → Playlist)**
   - **Aggregation (Playlist → Songs)**
5. **Draw the class diagram using tools like:**
   - **Azure DevOps**
   - **Draw.io**
   - **StarUML**

6B. Sequence Diagram:

1. **Identify actors:**
    - **User**
2. **Identify objects:**
    - **UI**
    - **Backend**
    - **Database**
3. **Define interaction flow:**
    - **User requests playlist creation**
    - **UI sends request to backend**
    - **Backend processes and stores data**
    - **Database returns confirmation**
4. **Draw sequence diagram showing:**
    - **Messages**
    - **Lifelines**
    - **Execution flow**





Result:

The Class Diagram and Sequence Diagram were successfully designed for the Music Playlist Batch Creator project, representing system structure and interaction flow.